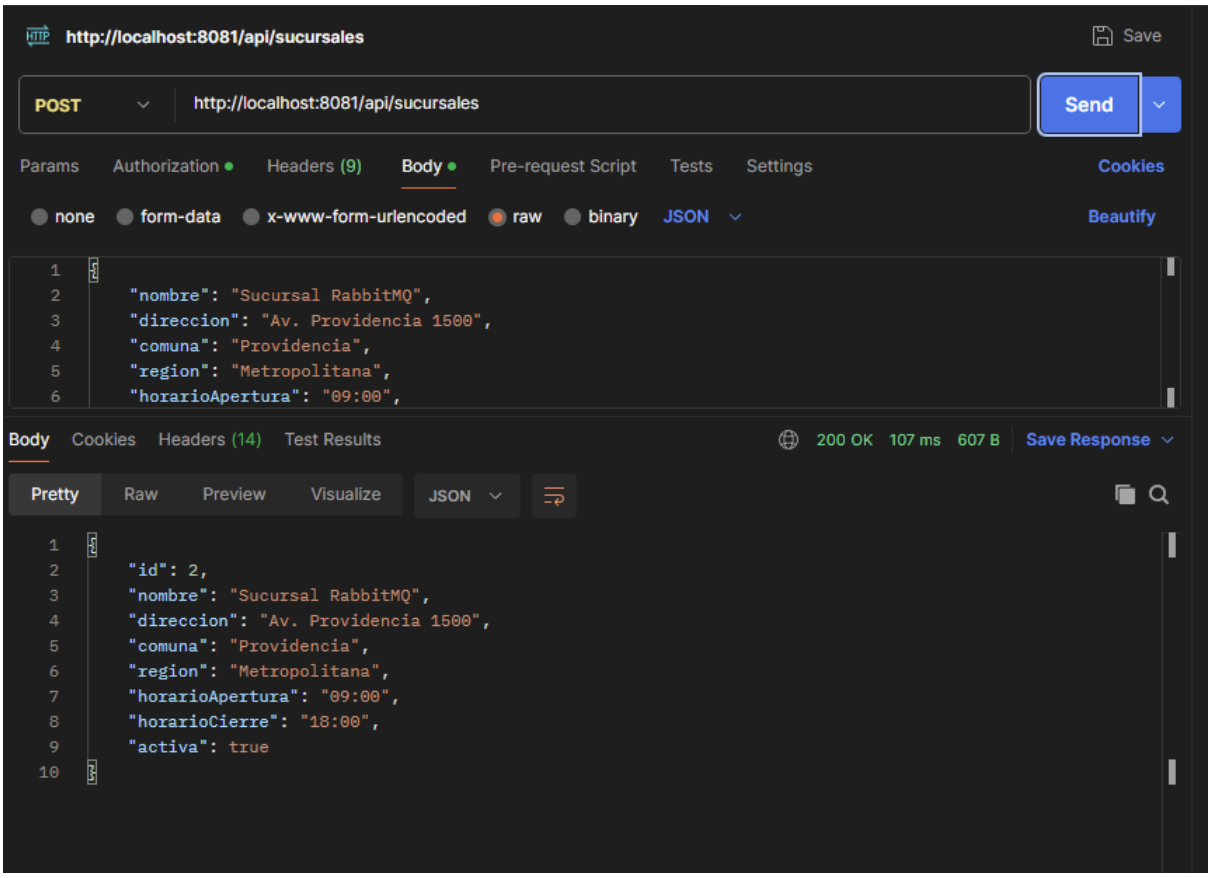
1.Postman creando una sucursal → 200 OK



2. Log "Evento SUCURSAL_CREADA enviado"

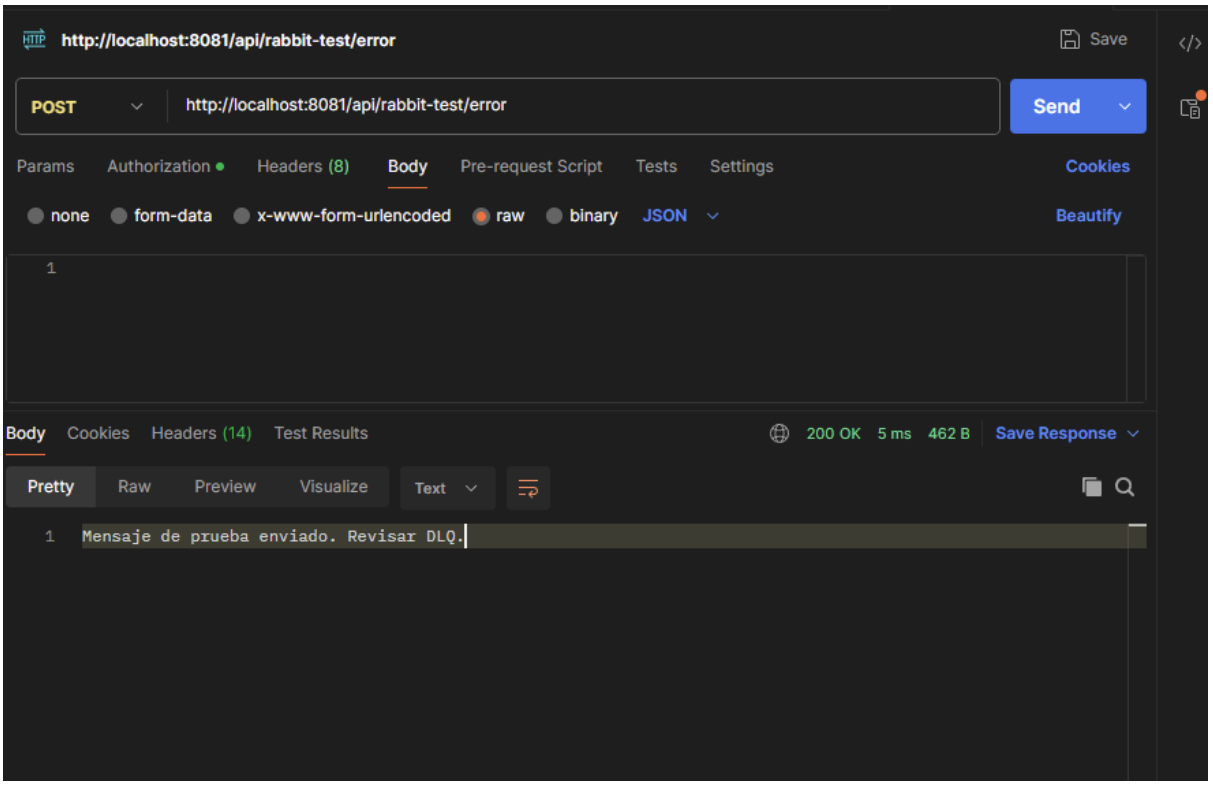
```
2026-10-26T15:44:18.049-03:00  WARN 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : NACK enviado. El mensaje ser| enviado a la DLQ.
Hibernate:
    insert
    into
        sucursales
        (activa, comuna, direccion, horario_apertura, horario_cierre, nombre, region)
    values
        (?, ?, ?, ?, ?, ?, ?)
2026-10-26T15:49:11.964-03:00  INFO 15444 --- [banco-cloud-sucursales] [0.0-8081-exec-5] c.b.s.r.producer.SucursalEventP
roducer   : Evento SUCURSAL_CREADA enviado a RabbitMQ: {
    "sucursalId": 2,
    "nombre": "Sucursal RabbitMQ",
    "comuna": "Providencia",
    "region": "Metropolitana",
    "evento": "SUCURSAL_CREADA"
}

2026-10-26T15:49:11.965-03:00  INFO 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : Mensaje recibido desde RabbitMQ: {
    "sucursalId": 2,
    "nombre": "Sucursal RabbitMQ",
    "comuna": "Providencia",
    "region": "Metropolitana",
    "evento": "SUCURSAL_CREADA"
}

2026-10-26T15:49:11.966-03:00  INFO 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : Mensaje procesado correctamente. ACK enviado.
```

3. Log "ACK enviado"

4. Postman /api/rabbit-test/error → 200 OK



5. Log del error

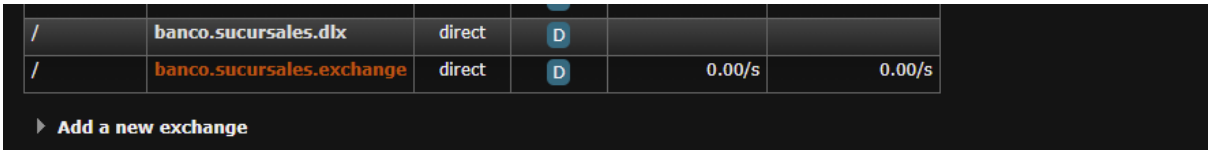
```
2026-10-26T15:49:11.966-03:00  INFO 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : Mensaje procesado correctamente. ACK enviado.
2026-10-26T15:50:40.364-03:00  WARN 15444 --- [banco-cloud-sucursales] [0.0-8081-exec-8] .w.s.m.s.DefaultHandlerExceptio
nResolver : Resolved [org.springframework.http.converter.HttpMessageNotReadableException: Required request body is missi
ng: public org.springframework.http.ResponseEntity<cl.bancocloud.sucursales.entity.Sucursal> cl.bancocloud.sucursales.co
ntroller.SucursalController.crear(cl.bancocloud.sucursales.entity.Sucursal)]
2026-10-26T15:50:49.016-03:00  INFO 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : Mensaje recibido desde RabbitMQ: {
    "sucursalId": 999,
    "nombre": "Prueba DLQ",
    "comuna": "Prueba",
    "region": "Prueba",
    "evento": "FORZAR_ERROR"
}

2026-10-26T15:50:49.016-03:00 ERROR 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : Error procesando mensaje: Error intencional para prueba de DLQ
2026-10-26T15:50:49.016-03:00  WARN 15444 --- [banco-cloud-sucursales] [ntContainer#0-1] c.b.s.r.consumer.SucursalEventC
onsumer   : NACK enviado. El mensaje ser├í enviado a la DLQ.
```

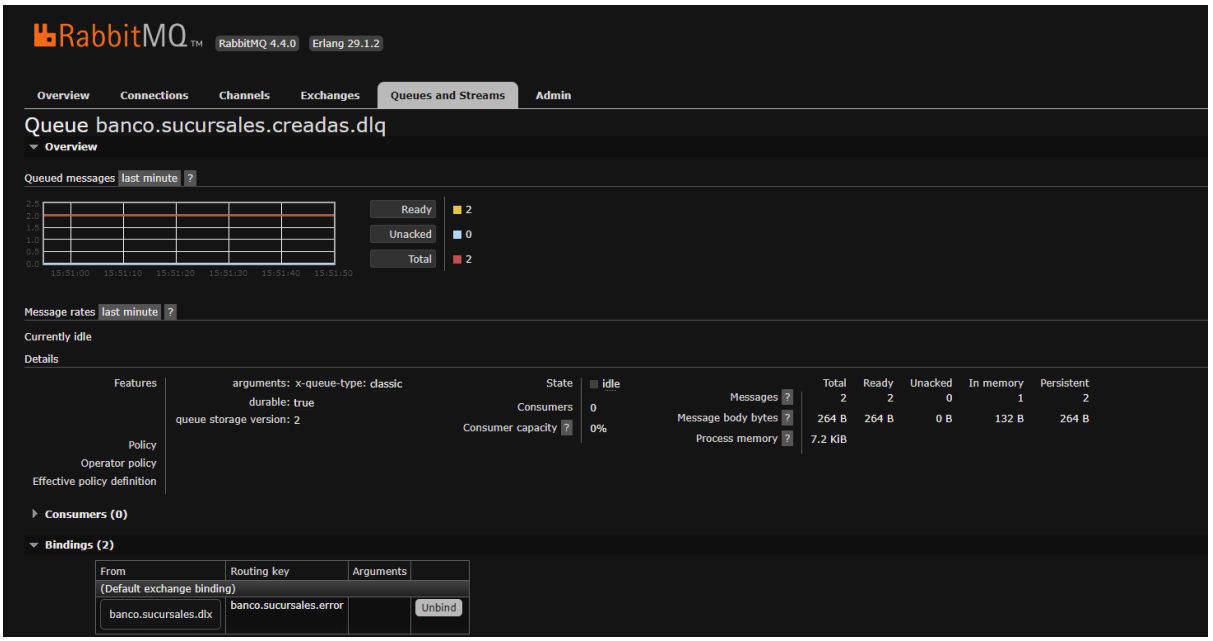
6. Log NACK

7. RabbitMQ DLQ → Ready: 1

8. RabbitMQ Exchanges

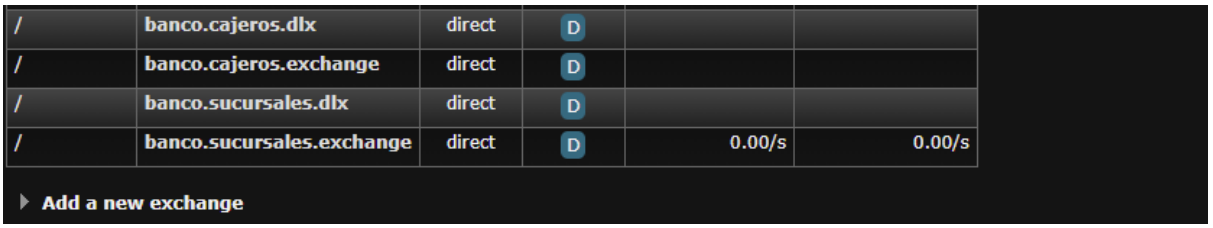


9. RabbitMQ Queues and Streams

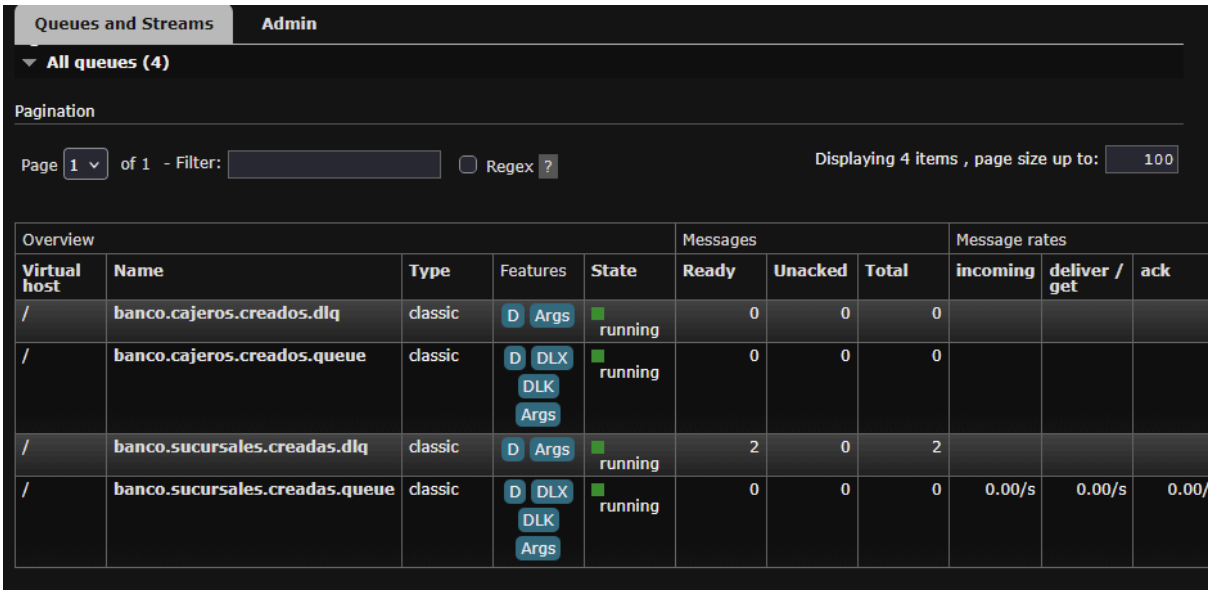


2cajeros:

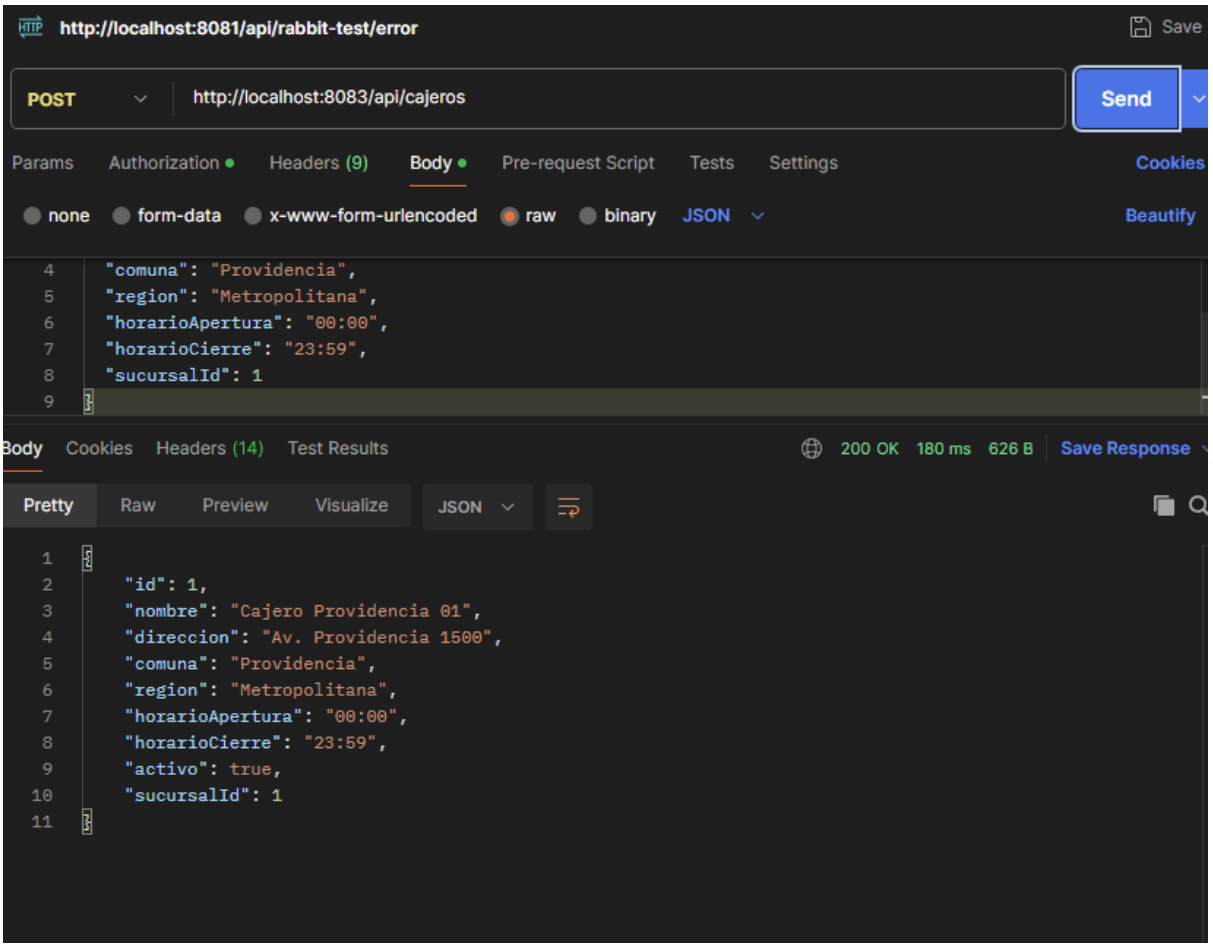
1



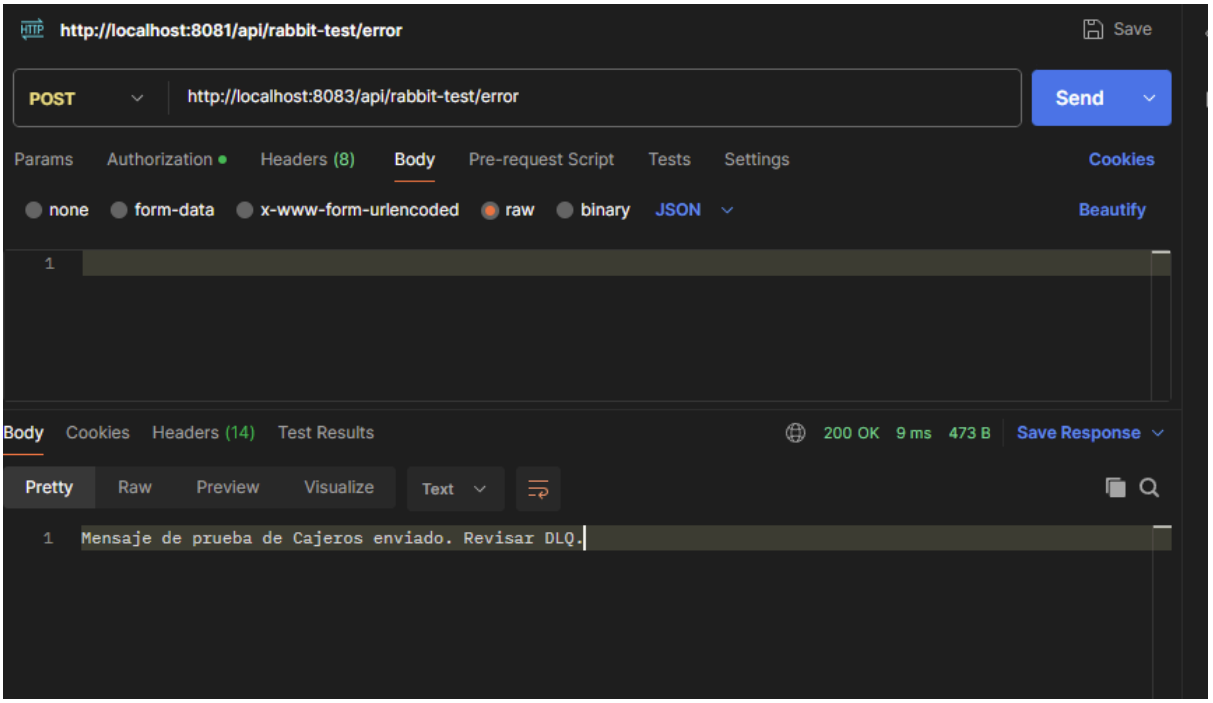
2



3


http://localhost:8081/api/rabbit-test/error
Save
POST
http://localhost:8083/api/cajeros
Send
Params
Authorization
Headers (9)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
"comuna": "Providencia",
"region": "Metropolitana",
"horarioApertura": "00:00",
"horarioCierre": "23:59",
"sucursalId": 1
Body
Cookies
Headers (14)
Test Results
200 OK
180 ms
626 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
"id": 1,
"nombre": "Cajero Providencia 01",
"direccion": "Av. Providencia 1500",
"comuna": "Providencia",
"region": "Metropolitana",
"horarioApertura": "00:00",
"horarioCierre": "23:59",
"activo": true,
"sucursalId": 1

4

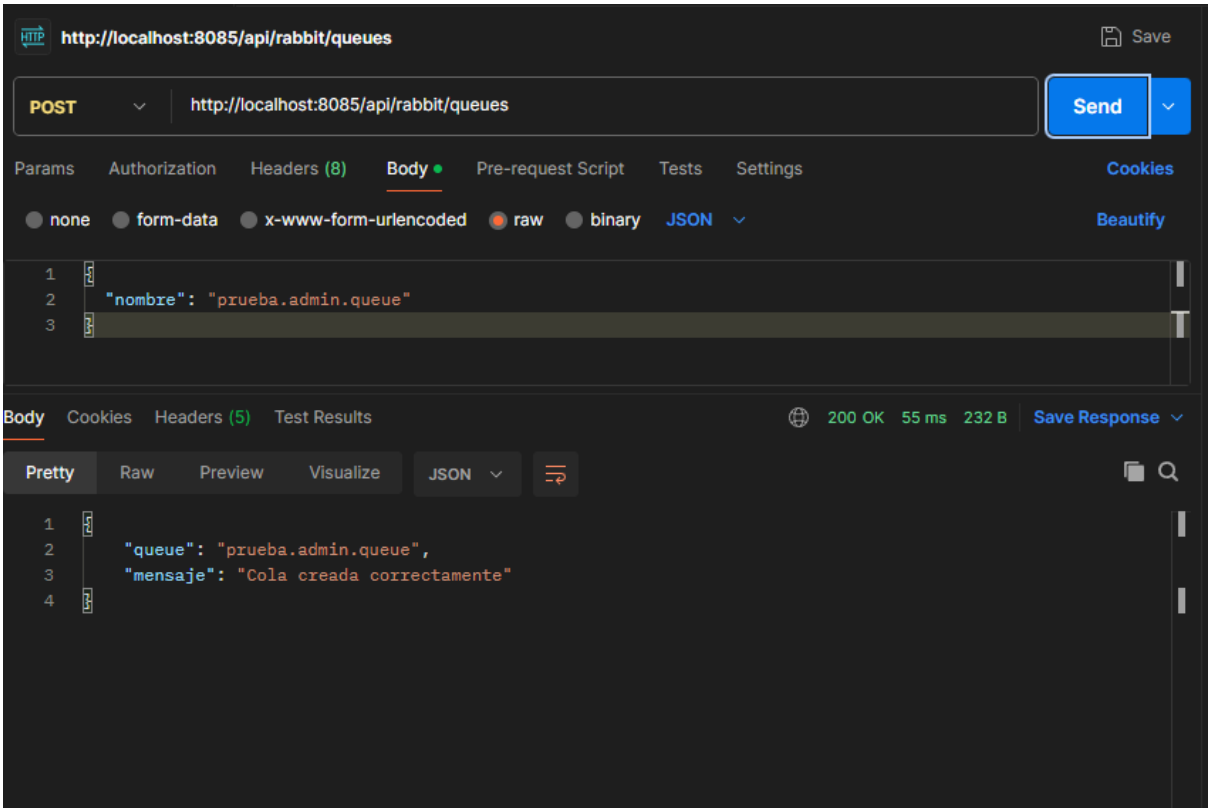

http://localhost:8081/api/rabbit-test/error
Save
POST
http://localhost:8083/api/rabbit-test/error
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
Body
Cookies
Headers (14)
Test Results
200 OK
9 ms
473 B
Save Response
Pretty
Raw
Preview
Visualize
Text
Mensaje de prueba de Cajeros enviado. Revisar DLQ.

5

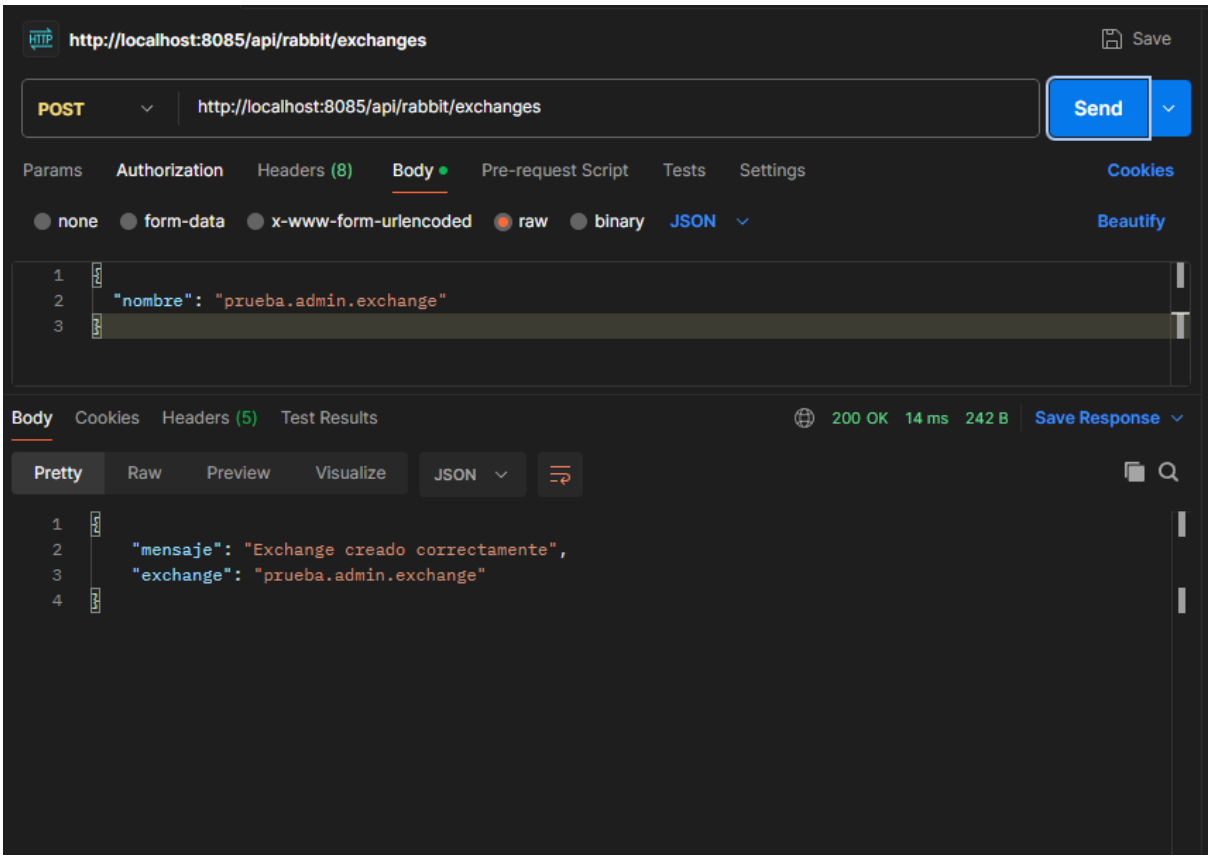
```
2026-10-26T16:34:39.016-03:00  INFO 8036 --- [banco-cloud-cajeros] [ntContainer#0-1] b.rabbit.consumer.CajeroEventConsum
er     : Mensaje de cajero recibido desde RabbitMQ: {
    "cajeroId": 999,
    "nombre": "Cajero prueba DLQ",
    "comuna": "Prueba",
    "region": "Prueba",
    "sucursalId": 999,
    "evento": "FORZAR_ERROR"
}

2026-10-26T16:34:39.016-03:00 ERROR 8036 --- [banco-cloud-cajeros] [ntContainer#0-1] b.rabbit.consumer.CajeroEventConsum
er     : Error procesando mensaje de cajero: Error intencional de Cajeros para prueba de DLQ
2026-10-26T16:34:39.018-03:00  WARN 8036 --- [banco-cloud-cajeros] [ntContainer#0-1] b.rabbit.consumer.CajeroEventConsum
er     : NACK enviado. Mensaje de cajero enviado a la DLQ.
```

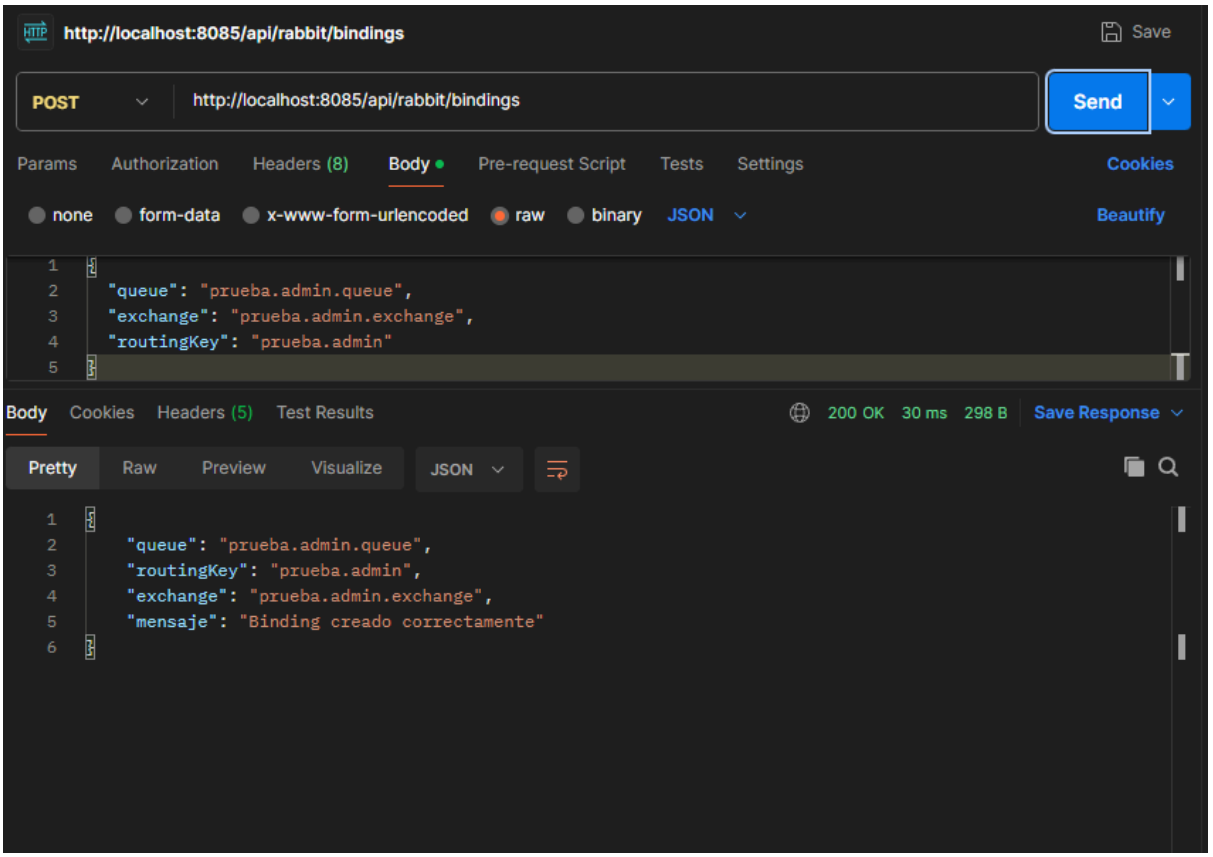
6



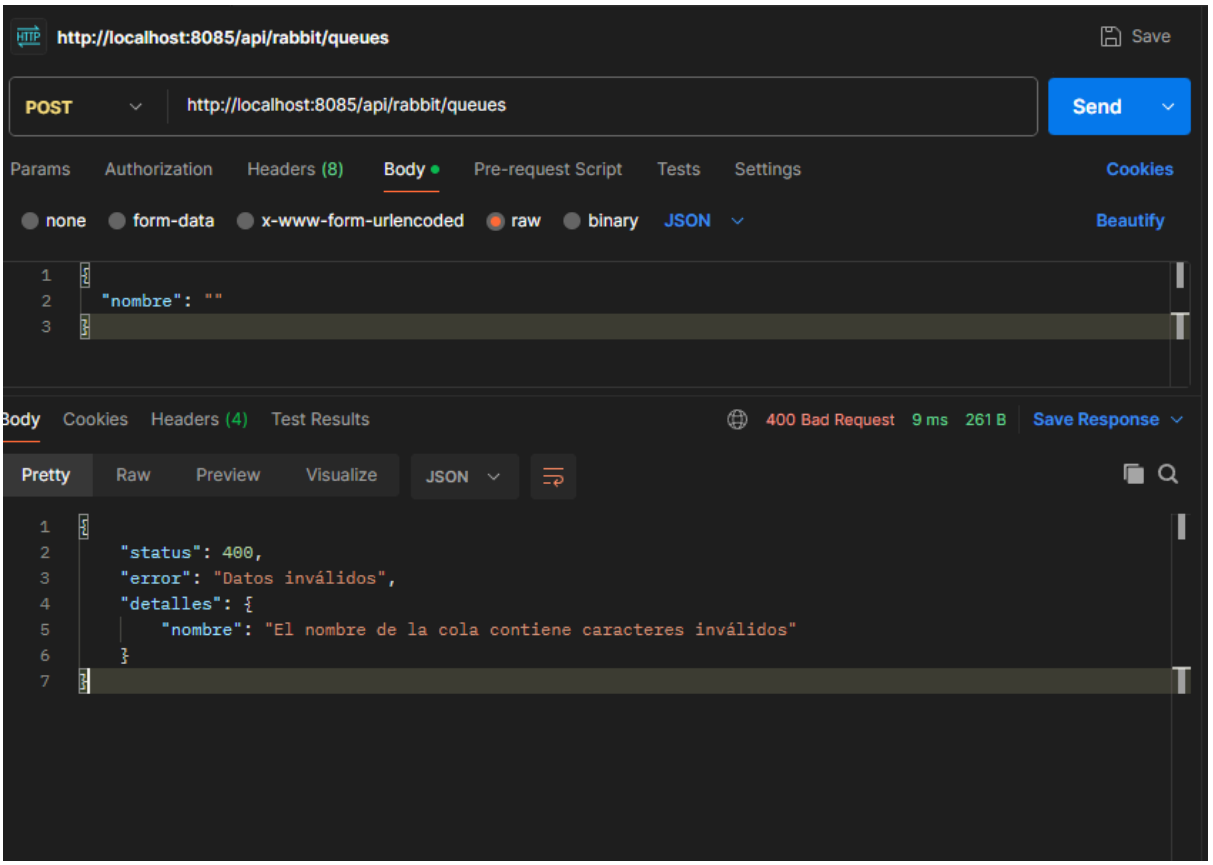
7


http://localhost:8085/api/rabbit/exchanges
Save
POST
http://localhost:8085/api/rabbit/exchanges
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
"nombre": "prueba.admin.exchange"
Body
Cookies
Headers (5)
Test Results
200 OK
14 ms
242 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
"mensaje": "Exchange creado correctamente",
"exchange": "prueba.admin.exchange"

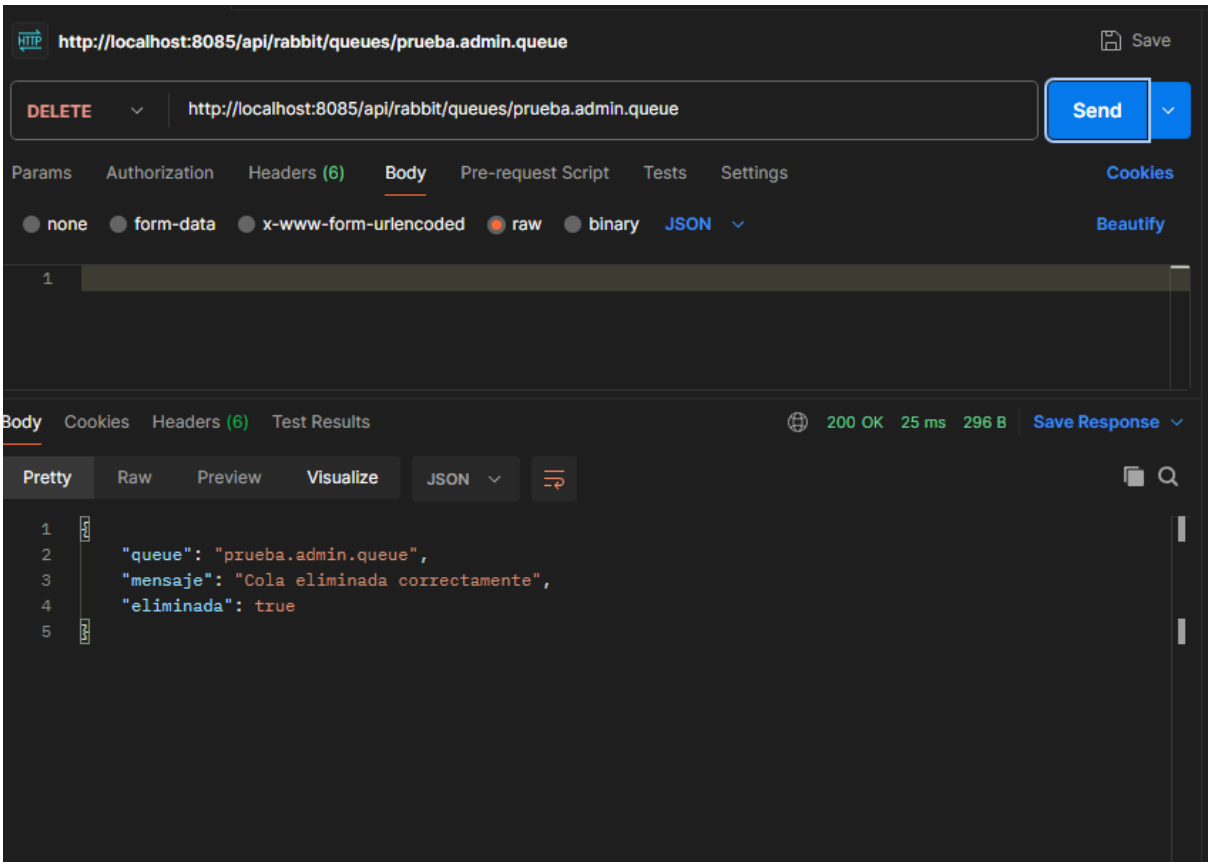
8


http://localhost:8085/api/rabbit/bindings
Save
POST
http://localhost:8085/api/rabbit/bindings
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
"queue": "prueba.admin.queue",
"exchange": "prueba.admin.exchange",
"routingKey": "prueba.admin"
Body
Cookies
Headers (5)
Test Results
200 OK
30 ms
298 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
"queue": "prueba.admin.queue",
"routingKey": "prueba.admin",
"exchange": "prueba.admin.exchange",
"mensaje": "Binding creado correctamente"

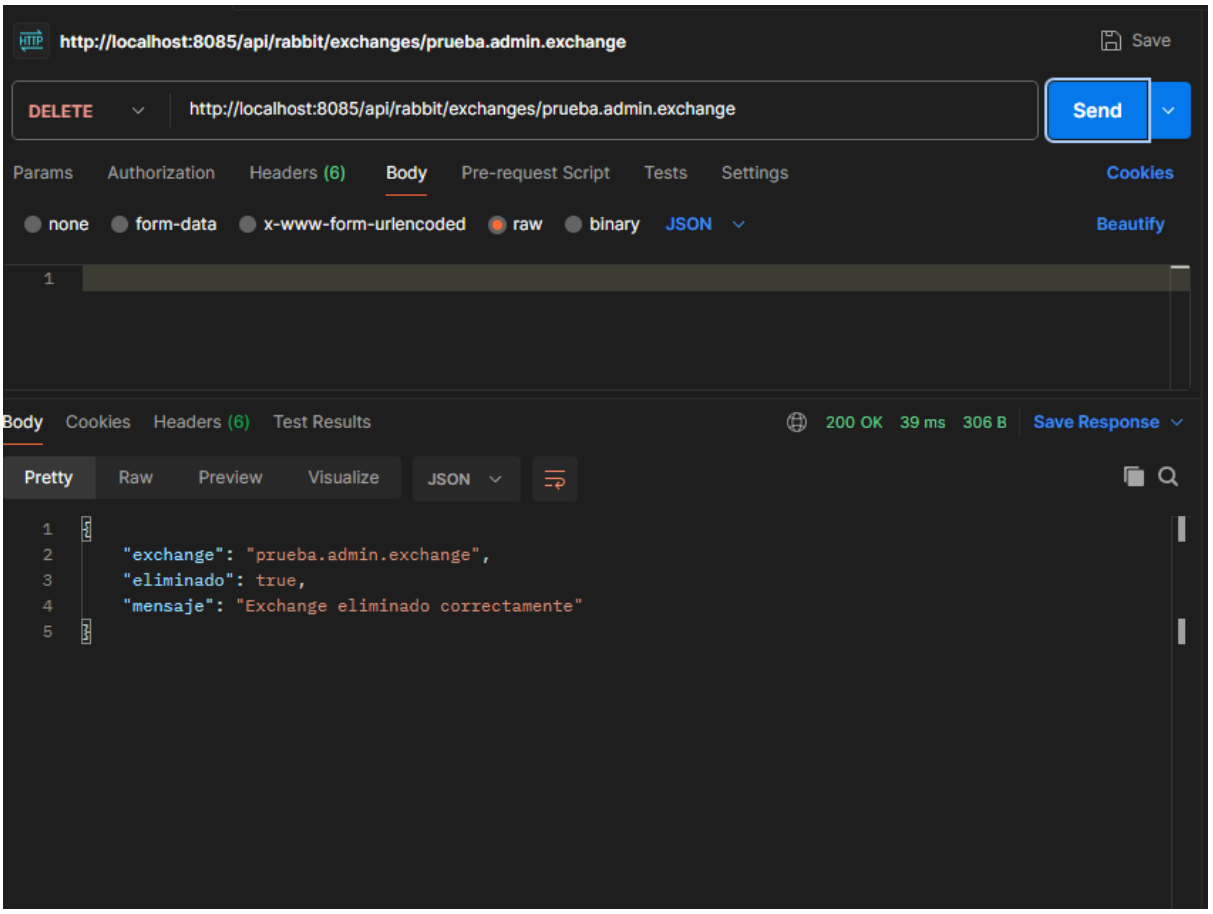
9


http://localhost:8085/api/rabbit/queues
Save
POST
http://localhost:8085/api/rabbit/queues
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
{
"nombre": ""
}
Body
Cookies
Headers (4)
Test Results
400 Bad Request
9 ms
261 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"status": 400,
"error": "Datos inválidos",
"detalles": {
"nombre": "El nombre de la cola contiene caracteres inválidos"
}
}

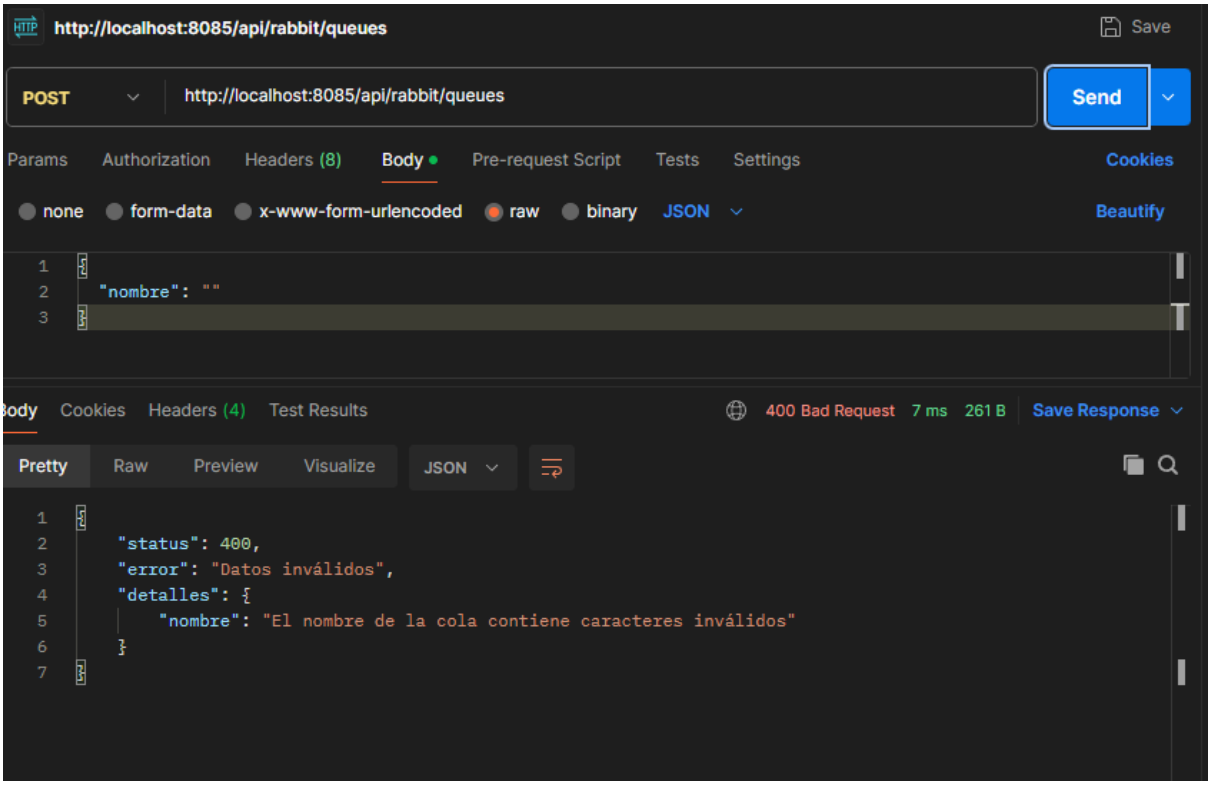
10


http://localhost:8085/api/rabbit/queues/prueba.admin.queue
Save
DELETE
http://localhost:8085/api/rabbit/queues/prueba.admin.queue
Send
Params
Authorization
Headers (6)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
Body
Cookies
Headers (6)
Test Results
200 OK
25 ms
296 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
{
"queue": "prueba.admin.queue",
"mensaje": "Cola eliminada correctamente",
"eliminada": true
}

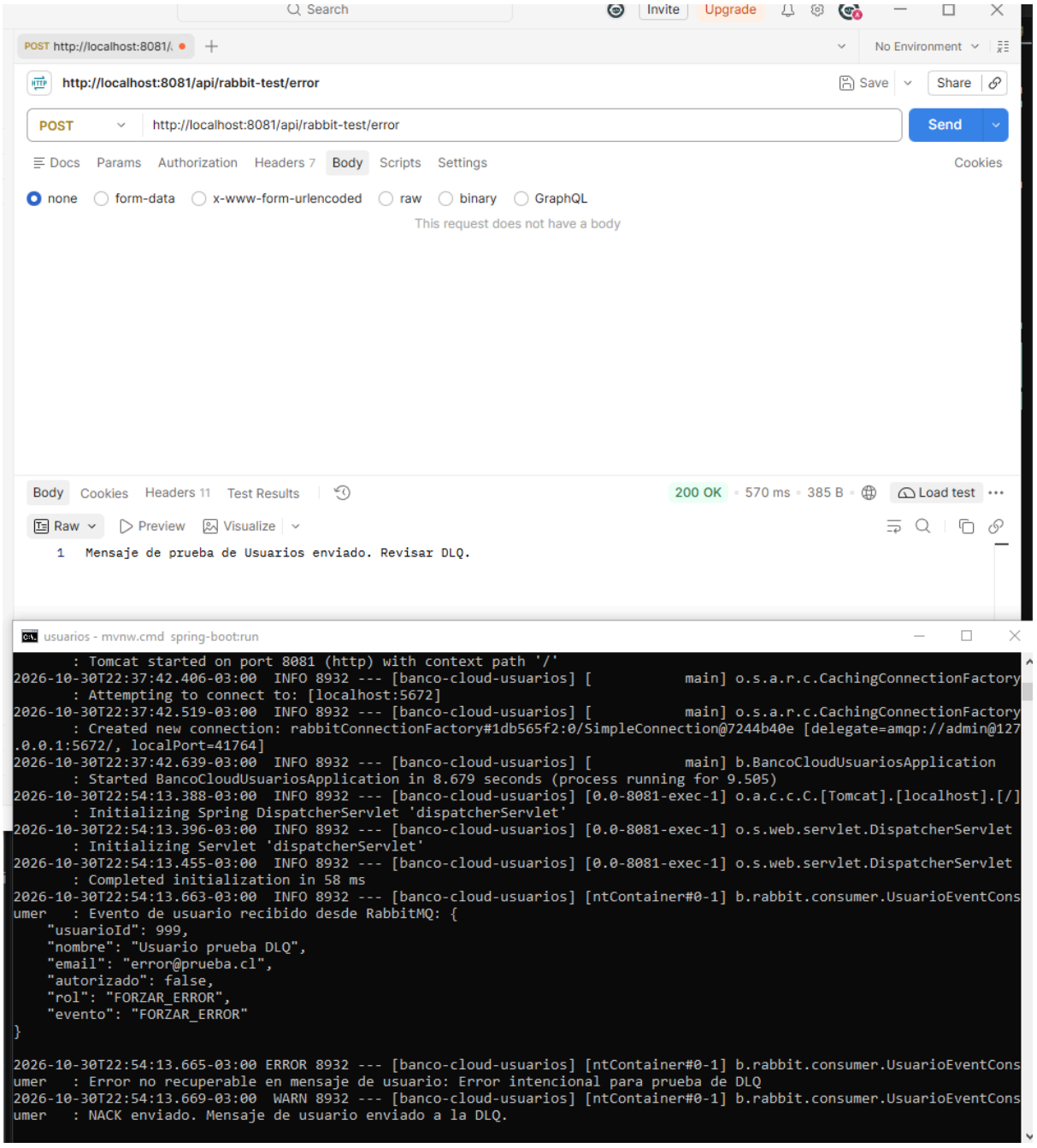
11


http://localhost:8085/api/rabbit/exchanges/prueba.admin.exchange
Save
DELETE
http://localhost:8085/api/rabbit/exchanges/prueba.admin.exchange
Send
Params
Authorization
Headers (6)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
1
Body
Cookies
Headers (6)
Test Results
200 OK
39 ms
306 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
1 {
2 "exchange": "prueba.admin.exchange",
3 "eliminado": true,
4 "mensaje": "Exchange eliminado correctamente"
5 }

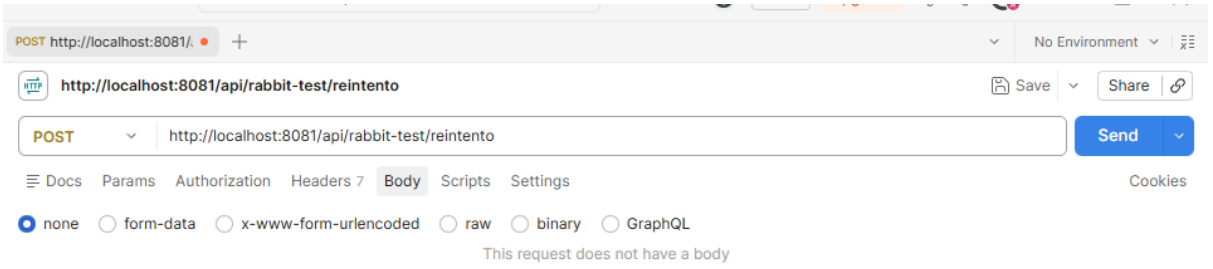
12

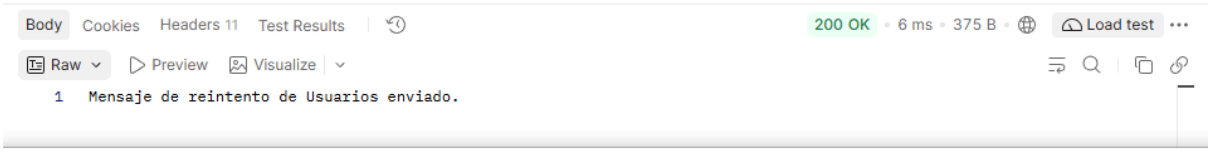

http://localhost:8085/api/rabbit/queues
Save
POST
http://localhost:8085/api/rabbit/queues
Send
Params
Authorization
Headers (8)
Body
Pre-request Script
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
JSON
Beautify
1 {
2 "nombre": ""
3 }
Body
Cookies
Headers (4)
Test Results
400 Bad Request
7 ms
261 B
Save Response
Pretty
Raw
Preview
Visualize
JSON
1 {
2 "status": 400,
3 "error": "Datos inválidos",
4 "detalles": {
5 "nombre": "El nombre de la cola contiene caracteres inválidos"
6 }
7 }

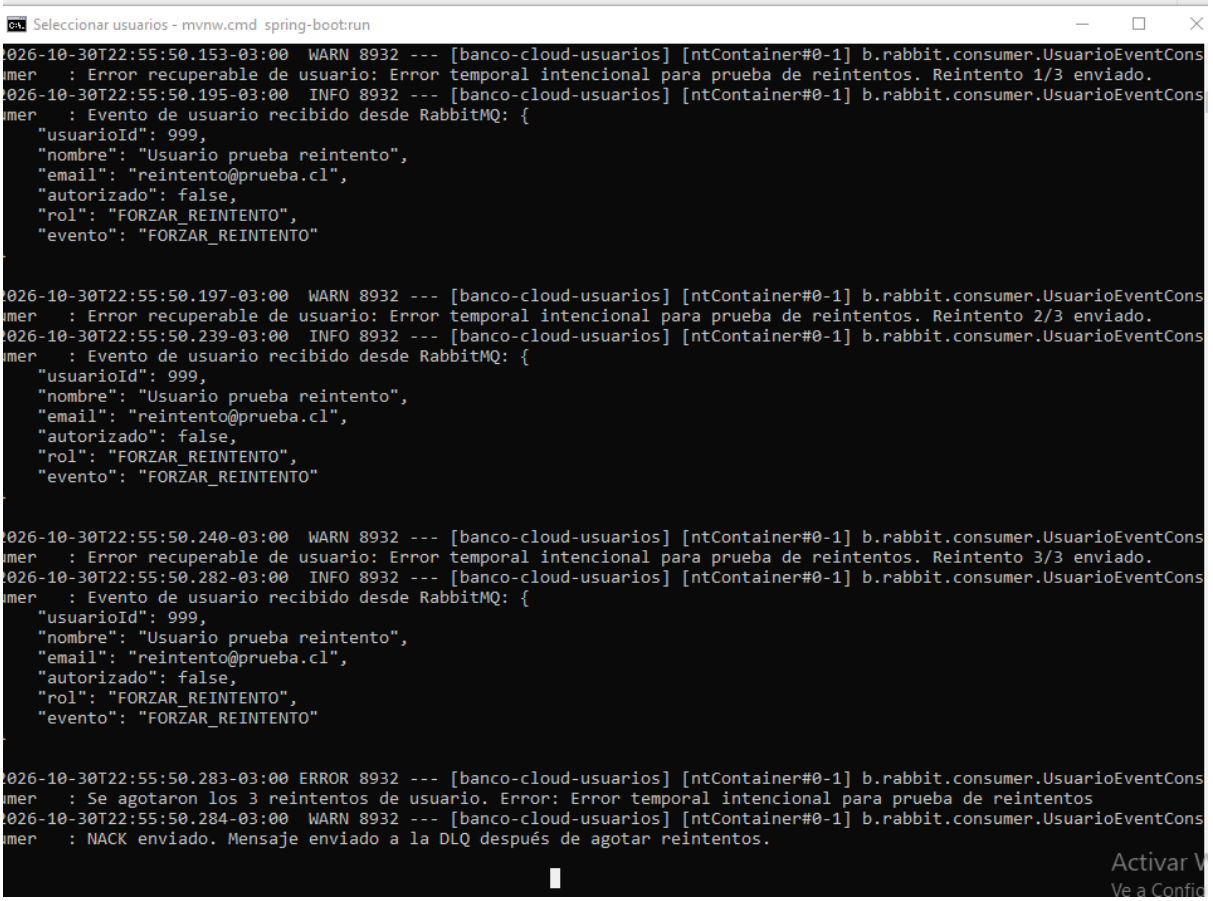
13


Search
Invite
Upgrade
No Environment
http://localhost:8081/api/rabbit-test/error
Save
Share
POST
http://localhost:8081/api/rabbit-test/error
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
This request does not have a body
Body
Cookies
Headers 11
Test Results
200 OK
570 ms
385 B
Load test
Raw
Preview
Visualize
1 Mensaje de prueba de Usuarios enviado. Revisar DLQ.
usuarios - mvnw.cmd spring-boot:run
: Tomcat started on port 8081 (http) with context path '/'
2026-10-30T22:37:42.406-03:00 INFO 8932 --- [banco-cloud-usuarios] [main] o.s.a.r.c.CachingConnectionFactory
: Attempting to connect to: [localhost:5672]
2026-10-30T22:37:42.519-03:00 INFO 8932 --- [banco-cloud-usuarios] [main] o.s.a.r.c.CachingConnectionFactory
: Created new connection: rabbitConnectionFactory#1db565f2:0/SimpleConnection@7244b40e [delegate=amqp://admin@127
.0.0.1:5672/, localPort=41764]
2026-10-30T22:37:42.639-03:00 INFO 8932 --- [banco-cloud-usuarios] [main] b.BancoCloudUsuariosApplication
: Started BancoCloudUsuariosApplication in 8.679 seconds (process running for 9.505)
2026-10-30T22:54:13.388-03:00 INFO 8932 --- [banco-cloud-usuarios] [0.0-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/]
: Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-10-30T22:54:13.396-03:00 INFO 8932 --- [banco-cloud-usuarios] [0.0-8081-exec-1] o.s.web.servlet.DispatcherServlet
: Initializing Servlet 'dispatcherServlet'
2026-10-30T22:54:13.455-03:00 INFO 8932 --- [banco-cloud-usuarios] [0.0-8081-exec-1] o.s.web.servlet.DispatcherServlet
: Completed initialization in 58 ms
2026-10-30T22:54:13.663-03:00 INFO 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Evento de usuario recibido desde RabbitMQ: {
"usuarioId": 999,
"nombre": "Usuario prueba DLQ",
"email": "error@prueba.cl",
"autorizado": false,
"rol": "FORZAR_ERROR",
"evento": "FORZAR_ERROR"
}
2026-10-30T22:54:13.665-03:00 ERROR 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Error no recuperable en mensaje de usuario: Error intencional para prueba de DLQ
2026-10-30T22:54:13.669-03:00 WARN 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : NACK enviado. Mensaje de usuario enviado a la DLQ.

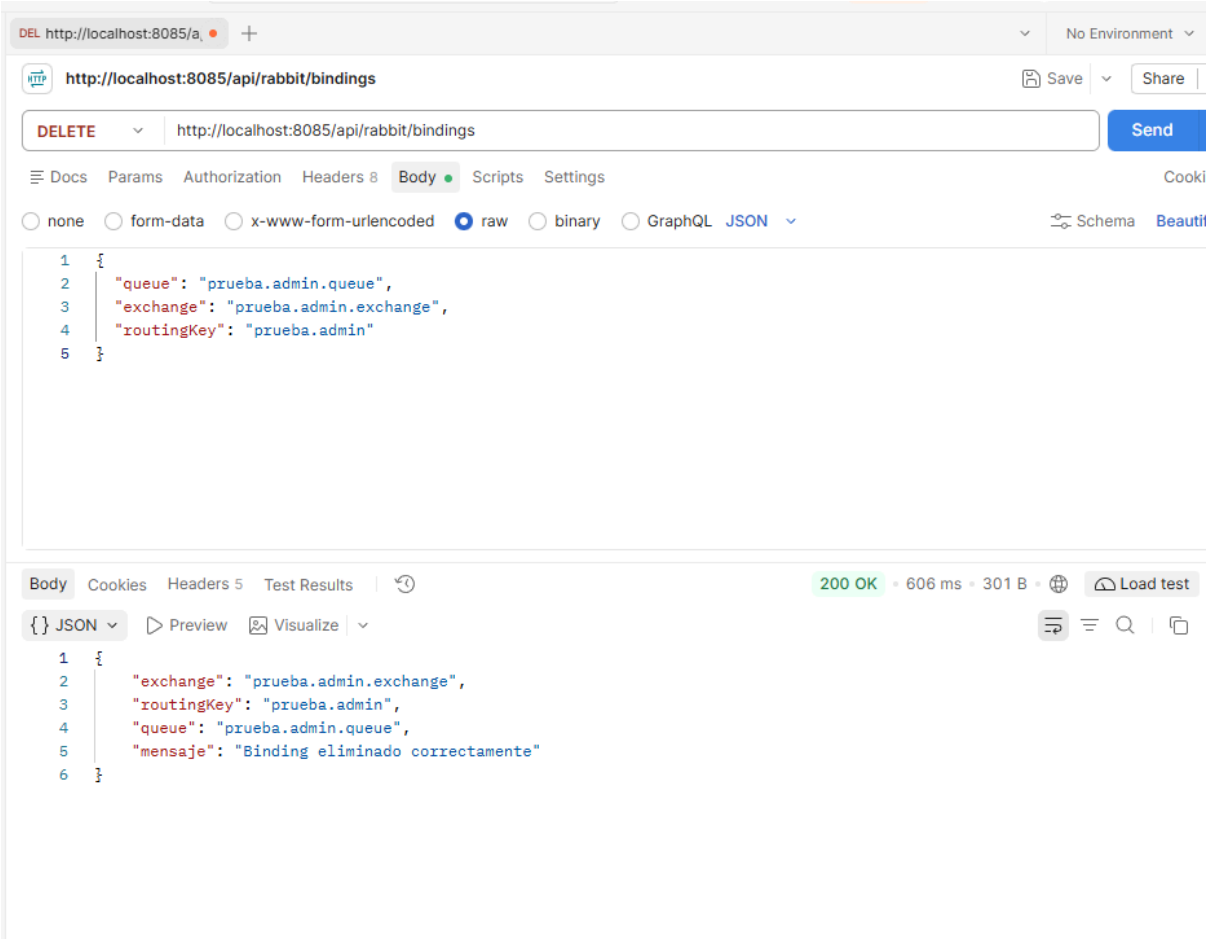
14


POST http://localhost:8081/
No Environment
http://localhost:8081/api/rabbit-test/reintento
Save
Share
POST
http://localhost:8081/api/rabbit-test/reintento
Send
Docs
Params
Authorization
Headers 7
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
This request does not have a body


Body
Cookies
Headers 11
Test Results
200 OK
6 ms
375 B
Load test
Raw
Preview
Visualize
1 Mensaje de reintento de Usuarios enviado.


Seleccionar usuarios - mvnw.cmd spring-boot:run
2026-10-30T22:55:50.153-03:00 WARN 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Error recuperable de usuario: Error temporal intencional para prueba de reintentos. Reintento 1/3 enviado.
2026-10-30T22:55:50.195-03:00 INFO 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Evento de usuario recibido desde RabbitMQ: {
"usuarioId": 999,
"nombre": "Usuario prueba reintento",
"email": "reintento@prueba.cl",
"autorizado": false,
"rol": "FORZAR_REINTENTO",
"evento": "FORZAR_REINTENTO"
2026-10-30T22:55:50.197-03:00 WARN 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Error recuperable de usuario: Error temporal intencional para prueba de reintentos. Reintento 2/3 enviado.
2026-10-30T22:55:50.239-03:00 INFO 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Evento de usuario recibido desde RabbitMQ: {
"usuarioId": 999,
"nombre": "Usuario prueba reintento",
"email": "reintento@prueba.cl",
"autorizado": false,
"rol": "FORZAR_REINTENTO",
"evento": "FORZAR_REINTENTO"
2026-10-30T22:55:50.240-03:00 WARN 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Error recuperable de usuario: Error temporal intencional para prueba de reintentos. Reintento 3/3 enviado.
2026-10-30T22:55:50.282-03:00 INFO 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Evento de usuario recibido desde RabbitMQ: {
"usuarioId": 999,
"nombre": "Usuario prueba reintento",
"email": "reintento@prueba.cl",
"autorizado": false,
"rol": "FORZAR_REINTENTO",
"evento": "FORZAR_REINTENTO"
2026-10-30T22:55:50.283-03:00 ERROR 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : Se agotaron los 3 reintentos de usuario. Error: Error temporal intencional para prueba de reintentos
2026-10-30T22:55:50.284-03:00 WARN 8932 --- [banco-cloud-usuarios] [ntContainer#0-1] b.rabbit.consumer.UsuarioEventCons
umer : NACK enviado. Mensaje enviado a la DLQ después de agotar reintentos.
Activar
Ve a Confi

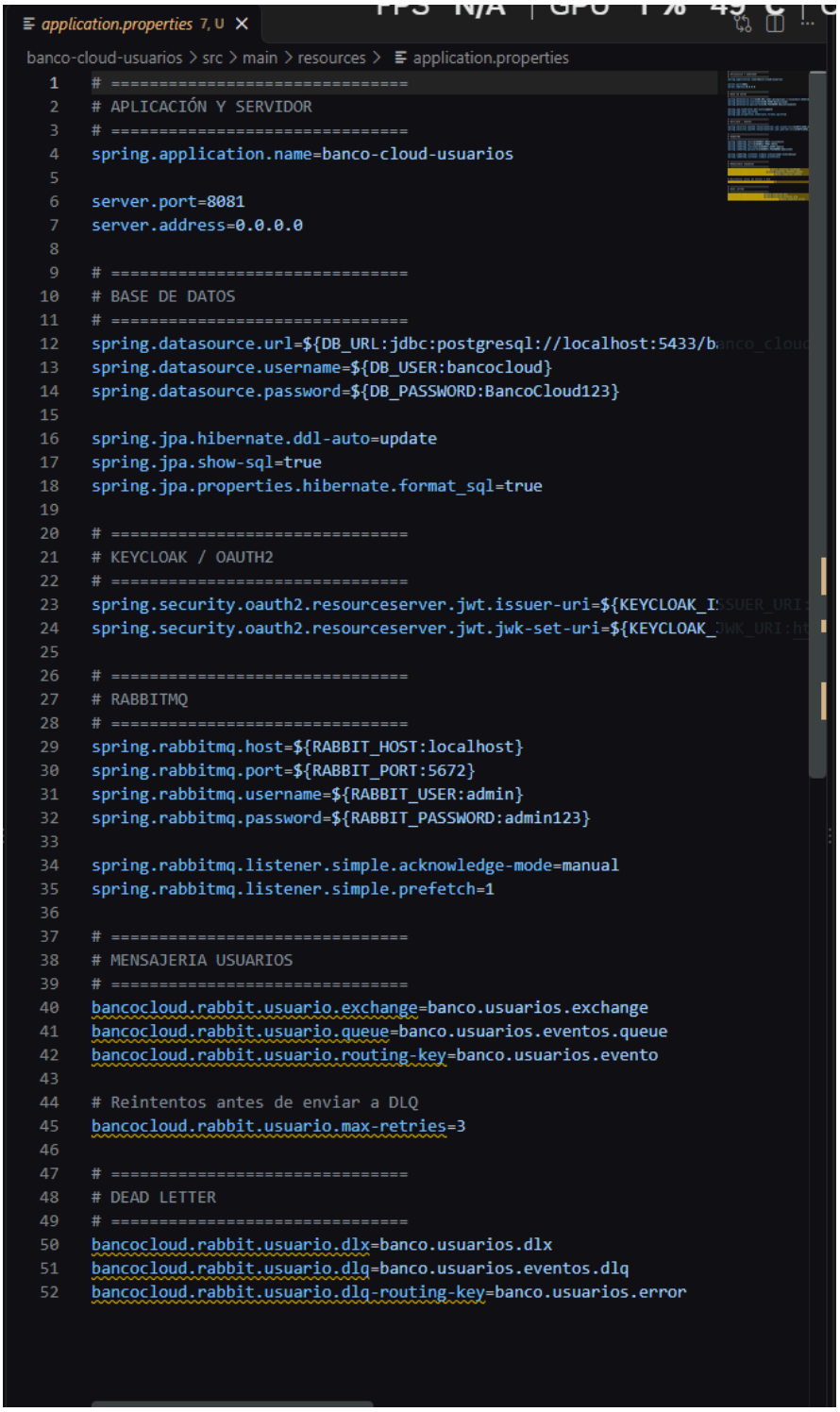
15


DEL http://localhost:8085/a
No Environment
http://localhost:8085/api/rabbit/bindings
Save
Share
DELETE
http://localhost:8085/api/rabbit/bindings
Send
Docs
Params
Authorization
Headers 8
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
{
"queue": "prueba.admin.queue",
"exchange": "prueba.admin.exchange",
"routingKey": "prueba.admin"
}
Body
Cookies
Headers 5
Test Results
200 OK
606 ms
301 B
Load test
JSON
Preview
Visualize
{
"exchange": "prueba.admin.exchange",
"routingKey": "prueba.admin",
"queue": "prueba.admin.queue",
"mensaje": "Binding eliminado correctamente"
}

16


RabbitAdminService.java M
ud-rabbit-admin > src > main > java > cl > bancocloud > rabbitadmin > service > RabbitAdminService.java
Usted, ayer | autores (TheTrup y otro más)
1 package cl.bancocloud.rabbitadmin.service;
TheTrup, hace 3
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.amqp.core.AmqpAdmin;
6 import org.springframework.amqp.core.Binding;
7 import org.springframework.amqp.core.BindingBuilder;
8 import org.springframework.amqp.core.DirectExchange;
9 import org.springframework.amqp.core.Queue;
10 import org.springframework.amqp.core.QueueBuilder;
11 import org.springframework.stereotype.Service;
12
Usted, ayer | autores (TheTrup y otro más)
13 @Service
14 public class RabbitAdminService {
15
16 private static final Logger log =
17 LoggerFactory.getLogger(
18 RabbitAdminService.class
19);
20
21 private final AmqpAdmin rabbitAdmin;
22
23 public RabbitAdminService(
24 AmqpAdmin rabbitAdmin
25) {
26 this.rabbitAdmin = rabbitAdmin;
27 }
28
29
30 // ==========================
31 // QUEUES
32 // ==========================
33
34 public void crearQueue(
35 String nombre
36) {
37
38 log.info(
39 "Intentando crear cola: {}",
40 nombre
41);
42
43 Queue queue =
44 QueueBuilder
45 .durable(nombre)
46 .build();
47
48 String resultado =
49 rabbitAdmin.declareQueue(queue);
50
51 log.info(
52 "Cola creada correctamente: {} - resultado: {}",
53 nombre,
54 resultado
55);


RabbitConfig.java U
banco-cloud-usuarios > src > main > java > banco_cloud_usuarios > config > RabbitConfig.java > Language
package banco_cloud_usuarios.config;
import org.springframework.amqp.core.Binding;
import org.springframework.amqp.core.BindingBuilder;
import org.springframework.amqp.core.DirectExchange;
import org.springframework.amqp.core.Queue;
import org.springframework.amqp.core.QueueBuilder;
import org.springframework.amqp.rabbit.annotation.EnableRabbit;
import org.springframework.beans.factory.annotation.Value;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
@Configuration
@EnableRabbit
public class RabbitConfig {
@Value("${bancocloud.rabbit.usuario.exchange}")
private String usuarioExchange;
@Value("${bancocloud.rabbit.usuario.queue}")
private String usuarioQueue;
@Value("${bancocloud.rabbit.usuario.routing-key}")
private String usuarioRoutingKey;
@Value("${bancocloud.rabbit.usuario.dlx}")
private String usuarioDlx;
@Value("${bancocloud.rabbit.usuario.dlq}")
private String usuarioDlq;
@Value("${bancocloud.rabbit.usuario.dlq-routing-key}")
private String usuarioDlqRoutingKey;
@Bean
public DirectExchange usuarioExchange() {
return new DirectExchange(
usuarioExchange,
true,
false
);
}
@Bean
public DirectExchange usuarioDeadLetterExchange() {
return new DirectExchange(
usuarioDlx,
true,
false
);
}
@Bean
public Queue usuarioQueue() {



application.properties 7, U

banco-cloud-usuarios > src > main > resources > application.properties

```
# ================================
# APLICACIÓN Y SERVIDOR
# ================================
spring.application.name=banco-cloud-usuarios

server.port=8081
server.address=0.0.0.0

# ================================
# BASE DE DATOS
# ================================
spring.datasource.url=${DB_URL:jdbc:postgresql://localhost:5433/b
spring.datasource.username=${DB_USER:bancocloud}
spring.datasource.password=${DB_PASSWORD:BancoCloud123}

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true

# ================================
# KEYCLOAK / OAUTH2
# ================================
spring.security.oauth2.resourceserver.jwt.issuer-uri=${KEYCLOAK_I
spring.security.oauth2.resourceserver.jwt.jwk-set-uri=${KEYCLOAK_

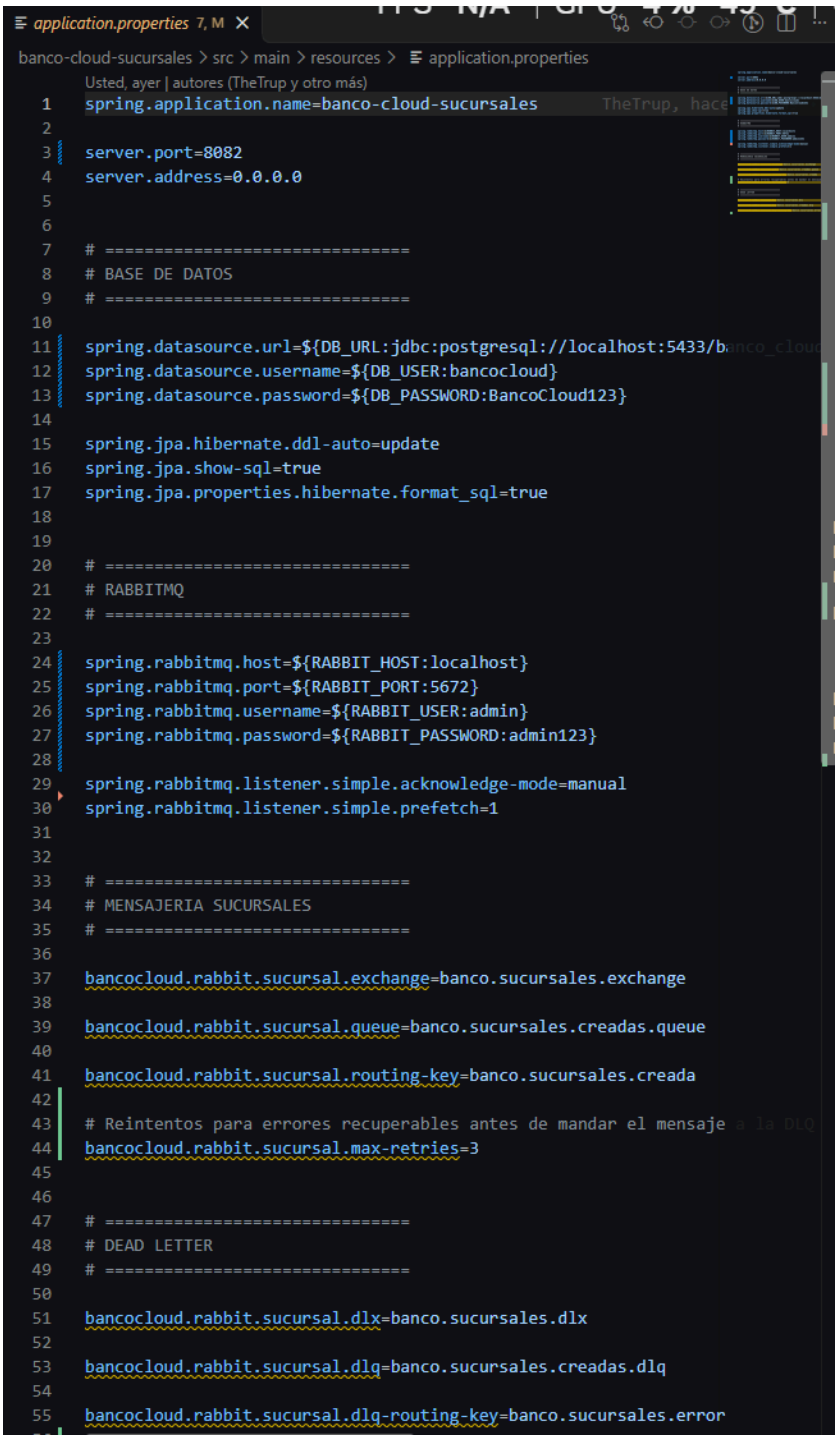
# ================================
# RABBITMQ
# ================================
spring.rabbitmq.host=${RABBIT_HOST:localhost}
spring.rabbitmq.port=${RABBIT_PORT:5672}
spring.rabbitmq.username=${RABBIT_USER:admin}
spring.rabbitmq.password=${RABBIT_PASSWORD:admin123}

spring.rabbitmq.listener.simple.acknowledge-mode=manual
spring.rabbitmq.listener.simple.prefetch=1

# ================================
# MENSAJERIA USUARIOS
# ================================
bancocloud.rabbit.usuario.exchange=banco.usuarios.exchange
bancocloud.rabbit.usuario.queue=banco.usuarios.eventos.queue
bancocloud.rabbit.usuario.routing-key=banco.usuarios.evento

# Reintentos antes de enviar a DLQ
bancocloud.rabbit.usuario.max-retries=3

# ================================
# DEAD LETTER
# ================================
bancocloud.rabbit.usuario.dlx=banco.usuarios.dlx
bancocloud.rabbit.usuario.dlq=banco.usuarios.eventos.dlq
bancocloud.rabbit.usuario.dlq-routing-key=banco.usuarios.error
```


application.properties 7, M
banco-cloud-sucursales > src > main > resources > application.properties
Usted, ayer | autores (TheTrup y otro más)
spring.application.name=banco-cloud-sucursales
server.port=8082
server.address=0.0.0.0
==============================
BASE DE DATOS
==============================
spring.datasource.url=${DB_URL:jdbc:postgresql://localhost:5433/b
spring.datasource.username=${DB_USER:bancocloud}
spring.datasource.password=${DB_PASSWORD:BancoCloud123}
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true
==============================
RABBITMQ
==============================
spring.rabbitmq.host=${RABBIT_HOST:localhost}
spring.rabbitmq.port=${RABBIT_PORT:5672}
spring.rabbitmq.username=${RABBIT_USER:admin}
spring.rabbitmq.password=${RABBIT_PASSWORD:admin123}
spring.rabbitmq.listener.simple.acknowledge-mode=manual
spring.rabbitmq.listener.simple.prefetch=1
==============================
MENSAJERIA SUCURSALES
==============================
bancocloud.rabbit.sucursal.exchange=banco.sucursales.exchange
bancocloud.rabbit.sucursal.queue=banco.sucursales.creadas.queue
bancocloud.rabbit.sucursal.routing-key=banco.sucursales.creada
Reintentos para errores recuperables antes de mandar el mensaje a la DLQ
bancocloud.rabbit.sucursal.max-retries=3
==============================
DEAD LETTER
==============================
bancocloud.rabbit.sucursal.dlx=banco.sucursales.dlx
bancocloud.rabbit.sucursal.dlq=banco.sucursales.creadas.dlq
bancocloud.rabbit.sucursal.dlq-routing-key=banco.sucursales.error